

CSE331 Assignment 1 Part A

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- There will be two parts in this assignment. Part A and Part B. Both need to be submitted together online in pdf format in a google form that will be published later.

- Deadline for entire Assignment 1 submission:

13 November, 2025, Thursday, 11:59 pm.

- Only one submission per group.
- The submitted document should have clear and legible scans.

1. (25 marks) Draw direct (No cross product rule) DFA and NFA accepting the following languages, $\Sigma = \{0, 1\}$:

- Has both even number of 0's and even number of 1's
- Binary representation of the string is divisible by 3 or 5
- Non-empty string such that there can be at most two consecutive 0's, and the pair should have at least one 1 in both its front and end
- Contains substring 101 at most two times
- Each 1 is followed by at least one 0 but cannot end with 100
- When substring 00 is found, substring 11 should exist as well and vice versa.
- The absolute value of the difference between the number of 0's and the number of 1's is odd.

2. (20 marks) Read the following description and draw DFA and NFA, $\Sigma = \{0, 1\}$:

- $L_1 = \{w \mid w = 1^n, n \text{ is a multiple of } 3\}$
- $L_2 = \{w \mid w = 0^m, m \text{ is 2 more than multiple of } 4\}$
- $L_3 = \{w \mid w \text{ contains } 0 \text{ in every position which is a multiple of } 3\}$
- $L_4 = \{w \mid w \text{ contains substring } 1011\}$

- $L_2 \cup L_3$
- $L_3 \cap L_4$
- $L_1 \circ L_2$
- $L_3 \oplus L_4$

3. (5 marks) Convert the following NFA into DFA, you need to show detailed steps:

